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EXAMPLE 10.12

A mass m moves with a speed v on a horizontal smooth surface and collides with a nearly massless spring whose spring constant is k. If the mass stops after collision, compute the maximum compression of the spring.

Solution

When the mass collides with the spring, from the law of conservation of energy "the loss in kinetic energy of mass is gain in elastic potential energy by spring".

Let x be the distance of compression of spring, then the law of conservation of energy

$$\frac{1}{2}m v^2 = \frac{1}{2}k x^2 \Rightarrow x = v\sqrt{\frac{m}{k}}$$

10.4.4 Oscillations of a simple pendulum in SHM and laws of simple pendulum

Simple pendulum

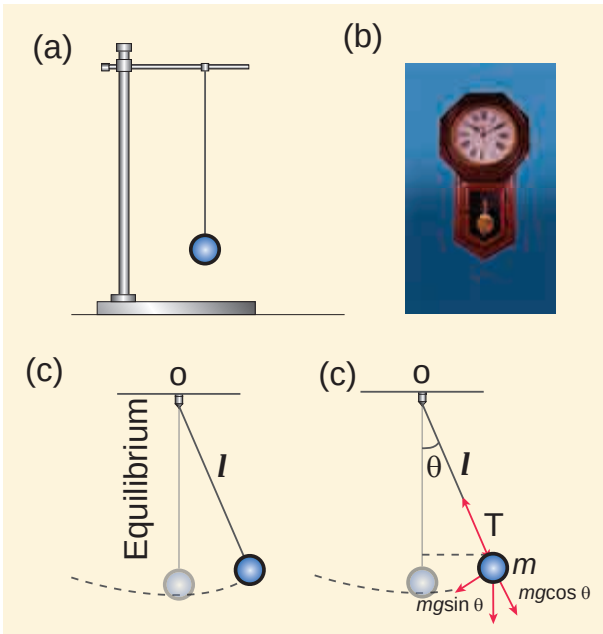


Figure 10.21 Simple pendulum

A pendulum is a mechanical system which exhibits periodic motion. It has a bob with mass m suspended by a long string (assumed to be massless and inextensible string) and the other end is fixed on a stand as shown in Figure 10.21 (a). At equilibrium, the pendulum does not oscillate and hangs vertically downward. Such a position is known as mean position or equilibrium position. When a pendulum is displaced through a small displacement from its equilibrium position and released, the bob of the pendulum executes to and fro motion. Let l be the length of the pendulum which is taken as the distance between the point of suspension and the centre of gravity of the bob. Two forces act on the bob of the pendulum at any displaced position, as shown in the Figure 10.21 (d),

(i) The gravitational force acting on the body ($\vec{F}=m\vec{g}$) which acts vertically downwards.

(ii) The tension in the string $\vec{T}$ which acts along the string to the point of suspension.

Resolving the gravitational force into its components:

a. **Normal component:** The component along the string but in opposition to the direction of tension, $F_{as} = mg\cos\theta$.

b. **Tangential component:** The component perpendicular to the string i.e., along tangential direction of arc of swing, $F_{ps} = mg\sin\theta$.

Therefore, The normal component of the force is, along the string,

$$T - W_{as} = m\frac{v^2}{l}$$

Here v is speed of bob

$$T - mg\cos\theta = m\frac{v^2}{l} \qquad (10.50)$$